IV. The Seam and the Mirror: Zero Divisors, the PT Automorphism, and the Orientation-Bearing Sector

Harald Rønning

Revision 1.1 — closed. (Rev. 1.1: the Revision 1.0 closure state was found to have regressed to a pre-closure draft during an unrelated numbering pass — caught and restored; the restoration folds in the later general-position confirmation from the Stage D work, which is new content beyond what Rev 1.0 itself closed on, hence the revision bump rather than a same-number restore.) A companion to *The Redistribution Operator* and the Phase 4 report. Every numerical claim below is two-implemented (independent code bases, results diffed); hand-derivations are tagged and were tested, not trusted; one operational definition is flagged as pending its primary source.

---

1. Charter

Phase 4 proved the orientation of the framework's chirality is protected from everything derived, leaving the settling stratum as sole author-candidate; the corpus's only named algebraic home for settling is the zero-divisor structure (Becoming §4) — the one structure of $\mathbb{S}$ that necessarily mixes both Cayley–Dickson halves. This note asks, and answers at the combinatorial level: **does the zero-divisor structure carry orientation information, and if so, exactly where?**

2. The tunnel lemma

[LEMMA, proven and verified] For a two-term zero divisor A, both diagonal blocks of $T\_A = L\_A^{T} L\_A$ are exactly flat — N(A)·Id on each half, by octonion norm-multiplicativity alone — so the three-band spectrum $\{0^4, 2^8, 4^4\}$ lives entirely in the **half-mixing cross-block** (singular values exactly {N(A)×4, 0×4}). The zero divisor's distinguishing content is quantitatively a *tunnel between the halves*, at maximal coupling strength σ = N(A). **[FACT, verified across four independent generators, including a general-position anchor]** The kernel is perfectly τ-balanced — every annihilated direction exactly half-even, half-odd — yet commutes with neither τ nor the mediator complex structure: a genuinely entangled subspace, orthogonal to every grading the corpus uses. Confirmed identically on three further, structurally distinct discrete generators beyond the classical pair, and on a genuine general-position anchor (rotated off the basis skeleton by a generic G_2 element, confirmed non-basis-aligned): flat blocks at machine precision, kernel dimension exactly 4, exact 0.5/0.5 balance, and nonzero [T_A, τ] in every case — the single-A worry is closed, not merely licensed by homogeneity.

[SCOPE REFINEMENT, added by Program ω, Stage C] "Commutes with neither τ nor the mediator complex structure" is generic, not universal. Across all 84 canonical diagonals, the kernel's J_0-invariance (equivalently, commuting with the mediator's own complex structure there) holds automatically for the pure-antilinear class and, more strikingly, as a genuine nontrivial fact for the mixed extreme class — exactly 24 diagonals in total, both extreme classes of the gate note's three-class taxonomy — and fails, as stated above, on all 60 generic diagonals. The four generators checked here were all generic; the exceptional set was not sampled by anything in this note and was found only once the dynamical construction made it visible. See the gate note, §6, for the mechanism (joint annihilation by L_A's J_0-linear and antilinear parts) and the full 84-diagonal confirmation.

3. The PT automorphism

[FACT, two-implemented] φ = diag(1,1,1,1,−1,−1,−1,−1) applied identically to both halves — fixing the real unit and the P-triple, negating the mediator and the X-triple — is an exact automorphism of $\mathbb{S}$ (zero violation over all 256 basis products), commuting with τ. **[INTERPRETATION, resting on verified facts]** In phase-space language φ is x → −x, p → +p with the seed complex structure conjugated to its negative ($\varphi L_{e_4} \varphi^{-1} = -L_{e_4}$, exact): the algebra's own **PT** operation. Since every gauge-side orientation descends from the mediator's sign (the J-identity chain), φ implements the mirror exchange.

4. Set-level neutrality and the orbit theorem

[FACT, two-implemented] φ maps the zero-divisor set to itself: the structure is PT-symmetric as a set — the protection theorem's reach extends across the seam. **[THM]** The four-group {id, τ, φ, τφ} is basis-diagonal, hence preserves every index-plane; τ exchanges each plane's two diagonals; all orbits are therefore exactly the diagonal-pairs, size 2. (This theorem resolved a canonicalization discrepancy in one implementation's first pass; the discrepancy's diagnosis and fix are in the development record.)

5. The de Marrais recovery and the character split

[FACT, two-implemented] The canonical two-term zero divisors number 84, on 42 index-planes with both diagonals active per plane — the 42 assessors — and the raw annihilation graph decomposes into **seven components of twelve with uniform degree 4**: the box-kites, recovered from annihilation relations alone with no classification imported. **[FACT, two-implemented]** The PT-character is **uniform within each kite** and splits the seven kites **3 + 4**: three kites wholly PT-even (every diagonal individually φ-fixed), four wholly PT-odd (every plane's diagonals φ-exchanged). No mixed kite exists.

[FACT, verified; operational definition flagged] Each kite omits exactly one Fano direction — the *same* index absent from its low and high halves — identifying the seven kites with the seven imaginary directions ("strut" in the missing-index reading, held one notch below full identification pending de Marrais's primary text). The correlation is exact, seven for seven: **a kite's PT-character equals the φ-parity of its strut** — P-strutted kites (e_1, e_2, e_3) are PT-even; kites strutted on the mediator and the X-triple (e_4; e_5, e_6, e_7) are PT-odd.

6. What this establishes

The orientation-bearing sector of the zero-divisor structure is exactly the four PT-odd box-kites — 48 diagonals on four components, where choosing a diagonal is choosing a PT-orientation — while the three P-strutted kites are provably mirror-blind. Two structural resonances, stated at their earned weights: **[FACT]** the axis of the split is the P/X distinction — the identical axis that constitutes the derived dynamics' entire internal content (A_sq, menu-fraction zero); the framework's flow-flavor and its seam-combinatorics share one axis, unarranged. **[INTERPRETATION]** the mirror-bearing kites are anchored on the position triple and the mediator ($\hbar$'s direction); the momentum-anchored kites carry nothing — orientation-information lives where the dynamics does.

The settling conjecture, sharpened to its most concrete form yet [CONJECTURE, standing condition unchanged]: settling selects the mirror iff its mechanism anchors in the PT-odd sector and breaks a diagonal symmetry there. "Does the state choose?" is now "which of seven addresses does the mechanism occupy, and does it split a pair?" — a finite question about a mapped landscape. Nothing here asserts the positive half; the Phase 4 verdict and its gating are unchanged.

7. Scope

All claims concern $\mathbb{S}$'s two-term zero divisors and the stated group action; general-position zero divisors (the full $V_2(\mathbb{R}^7)$ locus) inherit the set-level facts by G_2-homogeneity but the kite combinatorics is specifically the basis-diagonal story. The strut identification awaits the primary source. The tunnel lemma's kernel probes were extended beyond the classical single-A pair to three further discrete generators and one genuine general-position anchor, all confirming identically (§2) — homogeneity's licensing is now backed by direct instances rather than standing alone. This note adds no mechanism, selects no orientation, and moves no Phase 4 conclusion; it maps the terrain the surviving conjecture must cross.

[LEDGER] Open: the primary-source strut confirmation; the diagonal-symmetry question within PT-odd kites (what, if anything, natively distinguishes a plane's two diagonals beyond PT itself); transport of the kite map through the bridge dictionary to the wall sector's four orientations (the last unbuilt span of C1); and the general-position extension of the character split.

Created in collaboration with Claude (Anthropic). Closed after hostile review; the targets that review concentrated on, resolved as recorded: §2's kernel generality (single-A probes, extended to four independent generators including general position), §5's strut reading (operational vs primary definition), §6's resonance tags, and §7's claim that no Phase 4 conclusion moves. All conclusions and any errors remain the author's responsibility.